

FFY2023 MAS Portfolio Summary

Overview

The FFY2023 portfolio for ISWS under the FEMA Cooperating Technical Partners (CTP) program contains **14 active projects** (15 files — one is a superseded version of ISWS 23-01 marked "do not use"). Projects span the full Risk MAP lifecycle from early-stage survey data acquisition through near-final preliminary FIRM production. The portfolio includes 2 programmatic SOWs (PM and COMS), 5 regulatory mapping/production projects (DFIRM/PMR through preliminary issuance), 5 data development projects, 1 hydraulics-only project, 1 outreach-coordination project, and 1 standalone survey. Total FEMA funding committed is approximately **\$4.91M** (excluding the amendment ISWS 23-02A, which carries no additional funds). The dominant themes are: (1) bringing previously funded data development efforts into preliminary FIRM production across central and southern Illinois counties, (2) advancing Phase 4 of the large multi-phase Kishwaukee watershed study, and (3) building the data pipeline for future regulatory mapping along the Embarras and Kaskaskia watersheds. Nearly every production MAS stops at preliminary issuance and requires a future-funded phase for Due Process and final FIRM.

Note on files: `ISWS_23-01-PM_SOW_POP_incorrect_dont_use.pdf` is a superseded version of the PM SOW with an incorrect Period of Performance. The corrected version (`ISWS_23-01-PM_SOW - POP_Fixed_by_GNH.pdf`) is the authoritative document.

Projects

ISWS 23-01: Program Management (Statewide Illinois)

- **Activity Type:** Program Management (PM)
- **Phase:** Ongoing programmatic — not tied to a single mapping project phase; supports all other ISWS CTP projects
- **Scope:** ISWS provides statewide flood risk mapping program management under the CTP program, including developing a joint State Business Plan with IDNR-OWR and maintaining the Coordinated Needs Management Strategy (CNMS) database for all Illinois flood mapping projects. Work also includes global program management (risk assessments, CTP performance metrics in quarterly reports) and maintaining programmatic QA/QC plans. This is a management overhead SOW only — it cannot produce FIRM maps.
- **Deliverables:** State/ISWS Business Plan; Global Program Management Plan; CNMS spatial database updates at each mapping lifecycle milestone (Discovery, Production, Preliminary, LFD); Programmatic QA/QC Plan and checklists
- **Period:** 09/01/2023 – 10/01/2025
- **Budget:** \$95,000 (Task 1 Business Plans: \$30K; Task 2 Program Mgmt: \$25K; Task 10 CNMS: \$30K; Task 11 QA/QC: \$10K)

- **Multi-year:** Yes — PM is recurring; this SOW explicitly supports all companion flood risk project MAS across the full lifecycle (Discovery → Production → Preliminary → LFD). If companion COMS SOW (23-02) is not funded, Global Outreach tasks would need to be absorbed here.
- **Recommended Next Steps:** Ensure Business Plan is submitted within 4 months of award (by ~Jan 2024). CNMS deliverables must be submitted at each lifecycle milestone for all companion FY23 projects. Renew PM SOW for FFY2024+.

ISWS 23-02: Community Outreach and Mitigation Strategies — Kaskaskia Watershed (Statewide + Kaskaskia Focus)

- **Activity Type:** Community Outreach and Mitigation Strategies (COMS)
- **Phase:** Ongoing programmatic outreach; Kaskaskia Watershed strategic planning is forward-looking/pre-mapping
- **Scope:** ISWS provides statewide community outreach, communication, and flood risk mitigation support under the CTP program. Major focus this cycle is a Kaskaskia Watershed strategic planning effort (three watershed roundtables, community assessment/prioritization tool, mitigation action plan) to inform future mapping and outreach priorities. Work also includes maintaining illinoisfloodmaps.org and a statewide community contact database, conducting NFIP 101 and MT-2/LOMR training, providing mitigation planning technical assistance to at least 4 communities, and CTP mentoring/best practices.
- **Deliverables:** Combined COMS Engagement & Business Plan; Kaskaskia watershed community assessment + prioritization report; updated website + contact database + video tutorials; training materials (NFIP 101 video, MT-2/LOMR); mitigation planning TA report; mentoring report + 1 best-practice write-up for FEMA CTP Collaboration Center
- **Period:** 09/01/2023 – 09/30/2024
- **Budget:** \$200,000 (Strategic Planning/Kaskaskia: \$55K; Communications/Outreach: \$80K; Training: \$20K; Mitigation Planning TA: \$25K; COMS Plan: \$10K; Mentoring: \$10K)
- **Multi-year:** Yes — strategic planning explicitly identifies priorities for future CTP years; Kaskaskia watershed prioritization feeds future H&H/mapping project selection
- **Recommended Next Steps:** Deliver COMS Engagement Plan by ~March 2024; use Kaskaskia prioritization output to inform future MAS scoping for Kaskaskia data development. Note: ISWS 23-14 (Kaskaskia Survey) is already funded — coordinate timing. Renew COMS for FFY2024.

ISWS 23-02A: COMS SOW Amendment — Language Revision Only

- **Activity Type:** COMS SOW Amendment (administrative modification to ISWS 23-02)
- **Phase:** N/A — this is a language-only amendment, not a new phase
- **Scope:** This amendment modifies two passages in Section 2.2 (Strategic Planning for Community Outreach and Engagement) of the original ISWS 23-02 SOW. The changes remove references to "inequities" and "equitable" approaches, substituting more neutral language about identifying "barriers" and developing "a better approach." All other terms, deliverables, timelines, and budget of ISWS 23-02 remain unchanged.
- **Deliverables:** No new deliverables; replaces specific language in the original SOW
- **Period:** Inherits ISWS 23-02 period (09/01/2023 – 09/30/2024); signed by ISWS 05/28/2025

- **Budget:** \$200,000 (unchanged; no additional funds)
- **Multi-year:** No — administrative amendment only
- **Recommended Next Steps:** Ensure the amended language version is treated as the operative SOW 23-02 going forward. No action beyond acknowledgment needed.

ISWS 23-03: Peoria County Countywide DFIRM — QR1/Open House

- **Activity Type:** Countywide Digital FIRM (DFIRM) regulatory map production
- **Phase:** Pre-Preliminary through Preliminary Issuance + CCO Meeting + Public Open House (stops before Due Process). This is the culminating production phase consolidating multiple prior data development MAS efforts.
- **Scope:** Produces a countywide updated DFIRM for Peoria County (18 communities, 120 revised panels) by incorporating Zone A studies (ISWS11-16), Zone AE studies (ISWS18-04 and ISWS20-03), an updated Spoon River re-delineation using new LiDAR (ISWS21-09), and the USACE Illinois River Flow Frequency Study with updated topographic data. A Draft FIRM Database is developed, regulatory products produced, Preliminary Map Products issued, and a CCO meeting and Public Open House are conducted. Due Process, final map production, and Flood Risk Products development are explicitly out of scope.
- **Deliverables:** FIRM Database draft + preliminary files; 120 Preliminary FIRM panels (AMP tool); FIS Report; Preliminary Issuance Letters; Preliminary SOMAs; CCO Meeting documentation; Open House documentation; MIP/earned value reporting
- **Period:** 09/01/2023 – 09/30/2026
- **Budget:** \$260,100 (FEMA only; \$0 partner leverage)
- **Multi-year:** Yes — this project explicitly consolidates four prior ISWS MAS phases (ISWS11-16, 18-04, 20-03, 21-09). Also references 14 levee systems in the study area requiring ongoing monitoring. Future phases (Due Process, Final Map) are not funded.
- **Recommended Next Steps:** Monitor levee status changes (particularly Greater Peoria Sanitary District LOMR) that could delay Preliminary release. After Open House/CCO, fund a follow-on MAS for Due Process (90-day appeal period) and Final Map Production/Distribution (LFD).

ISWS 23-04: McHenry County Nippersink PMR — QR1/Open House

- **Activity Type:** Physical Map Revision (PMR) — limited panel
- **Phase:** Pre-Preliminary through Preliminary Issuance + CCO Meeting + Public Open House (stops before Due Process)
- **Scope:** Incorporates previously developed Nippersink Creek and tributary engineering studies (from ISWS15-01 and ISWS19-09) into the McHenry County FIS and FIRM, upgrading the FIS to current FEMA specifications. Covers Nippersink Creek, North Branch Nippersink Creek, and Spring Creek tributaries near Village of Spring Grove. Produces 15 revised FIRM panels, distributes Preliminary Map Products, and conducts a CCO meeting and Public Open House. Does not advance through Due Process or final map production.
- **Deliverables:** Draft FIRM Database; 15 Preliminary FIRM panels (AMP tool); FIS Report; Preliminary Issuance Letters; Preliminary SOMAs; CCO Meeting + Open House documentation; MIP reporting
- **Period:** 09/01/2023 – 09/30/2026

- **Budget:** \$252,100 (FEMA only; \$0 partner leverage)
- **Multi-year:** Yes — two prior data development phases (ISWS15-01, ISWS19-09) feed this production phase. Also incorporates leveraged FEMA Region V Zone A contractor studies. Due Process and final map production not funded.
- **Recommended Next Steps:** Post-Open House/CCO, fund a follow-on MAS for Due Process (90-day appeal, appeals resolution) and Final Map Production/LFD issuance to complete the Nippersink PMR regulatory lifecycle.

ISWS 23-05: McDonough County Countywide DFIRM — QR2/Open House

- **Activity Type:** Countywide Digital FIRM (DFIRM) regulatory map production
- **Phase:** Preliminary Map Production (QR2 stage — further along than the QR1 projects; production is in a more mature state at time of award). Produces and distributes 43 Preliminary FIRM panels with CCO meeting and Open House. Stops before Due Process.
- **Scope:** Incorporates Zone A and Zone AE engineering studies previously completed under MAS ISWS19-04 into a countywide McDonough County DFIRM covering 12 communities across 43 FIRM panels. Produces and distributes Preliminary Map Products and hosts a CCO meeting and Public Open House. Due Process, Final Map Production, and Flood Risk Products development are out of scope.
- **Deliverables:** 43 Preliminary FIRM panels; FIS Report; Preliminary Issuance Letters; Preliminary SOMAs; CCO Meeting + Open House documentation; MIP/earned value reporting
- **Period:** 09/01/2023 – 09/30/2026
- **Budget:** \$206,360 (FEMA only; \$0 partner leverage)
- **Multi-year:** Yes — builds on prior data development (ISWS19-04). Three non-accredited levees present (Natural Valley path forward). Due Process and final map production not funded.
- **Recommended Next Steps:** Being at QR2 at award likely means Preliminary issuance is nearer-term than QR1 projects. Prioritize funding a follow-on MAS for Due Process and Final Map Production/LFD, as this project may be ready for that next phase earliest among the FFY23 production projects.

ISWS 23-06: Edwards County Countywide DFIRM — QR1/Open House

- **Activity Type:** Countywide Digital FIRM (DFIRM) regulatory map production
- **Phase:** Pre-Preliminary (Draft FIRM Database) through Preliminary Issuance + CCO Meeting + Public Open House. Stops before Due Process.
- **Scope:** Incorporates Zone A and Zone AE engineering studies from MAS ISWS17-03 and ISWS18-03 into a countywide Edwards County DFIRM covering 6 communities across 17 FIRM panels. Produces and distributes Preliminary Map Products and hosts a CCO meeting and Public Open House. **Coordinated with Wabash County (ISWS 23-07):** Bonpas Creek forms the shared boundary — any substantial comments on that stream would affect both projects' budgets and schedules.
- **Deliverables:** Draft FIRM Database; 17 Preliminary FIRM panels; FIS Report; Preliminary Issuance Letters; Preliminary SOMAs; CCO Meeting + Open House documentation
- **Period:** 09/01/2023 – 09/30/2026
- **Budget:** \$205,500 (FEMA only; \$0 partner leverage)

- **Multi-year:** Yes — prior phases ISWS17-03 and ISWS18-03. Due Process and final map production not funded. Non-accredited levee (NLD 1505001152) mapped as Natural Valley.
- **Recommended Next Steps:** Maintain tight coordination with ISWS 23-07 (Wabash County) throughout — especially for Bonpas Creek comment resolution. Fund a joint or coordinated follow-on MAS for Due Process and Final Map Production covering both Edwards and Wabash Counties.

ISWS 23-07: Wabash County Countywide PMR — QR1/Open House

- **Activity Type:** Countywide Physical Map Revision (PMR) — incorporating studies + levee status update
- **Phase:** Pre-Preliminary (Draft FIRM Database) through Preliminary Issuance + CCO Meeting + Public Open House. Stops before Due Process.
- **Scope:** Updates the Wabash County effective DFIRM by incorporating Zone A and Zone AE engineering studies (from ISWS18-03) and updating Mt. Carmel Levee accreditation status to "Not Accredited" (results from LAMP project ISWS17-08). Covers 5 communities across 26 revised panels. Produces and distributes Preliminary Map Products and hosts one CCO meeting and one Open House. Eight levee systems in the study area are all proposed for Natural Valley mapping.
- **Deliverables:** Draft FIRM Database; 26 Preliminary FIRM panels; FIS Report; Preliminary Issuance Letters; Preliminary SOMAs; CCO Meeting + Open House documentation
- **Period:** 09/01/2023 – 09/30/2026
- **Budget:** \$237,540 (FEMA only; \$0 partner leverage)
- **Multi-year:** Yes — prior phases ISWS18-03 (data development) and ISWS17-08 (LAMP). Eight non-accredited levees. Due Process and final map production not funded. Bonpas Creek coordination risk shared with Edwards County (ISWS 23-06).
- **Recommended Next Steps:** Run Wabash County and Edwards County (23-06) mapping milestones in parallel, particularly for Bonpas Creek. After Open House/CCO, fund a coordinated follow-on MAS for Due Process and Final Map Production covering both counties. Engage levee owners early given 8 non-accredited systems.

ISWS 23-08: Kishwaukee Watershed Phase 4 Data Development (Winnebago/Boone/Ogle Counties)

- **Activity Type:** Data Development (Hydrologic and Hydraulic analysis + floodplain re-delineation)
- **Phase:** Phase 4 of the multi-phase Kishwaukee watershed study. Scope includes new Zone A hydrologic and hydraulic analysis in Winnebago, Boone, and Ogle Counties (~401 net NVUE miles) plus Zone AE floodplain re-delineations for valid effective streams using 2018 LiDAR. Goal is KDP2 submission.
- **Scope:** Develops new Zone A H&H data for approximately 285 miles of approximate streams and 83 miles of new/unmapped streams, plus 33 miles of new detailed Zone AE analysis in Winnebago, Boone, and Ogle Counties. Also re-delineates five Zone AE effective stream reaches (Beaver Creek, Piscasaw Creek, Coon Creek, Mosquito Creek, Tributary D — totaling ~44 miles) using updated LiDAR without re-modeling. Standard community outreach milestones (Project Initiation Meeting, Flood Risk Review Meeting, 30-day review periods) are included.

- **Deliverables:** Hydrology data/spatial files; Hydraulics simulations/FIS profiles/floodway data; Floodplain re-delineation spatial files for 5 Zone AE streams; independent QA/QC reports for all; Event Data Capture/outreach documentation; MIP monthly reporting
 - **Period:** 09/01/2023 – 09/30/2026
 - **Budget:** \$1,697,002 (FEMA only; \$0 partner leverage) — by far the largest single project in the portfolio
 - **Multi-year:** Yes — this is Phase 4 of an ongoing multi-phase watershed effort; prior detailed study for Kishwaukee River mainstem was previously completed. Future phases for Draft FIRM Database, Preliminary Production, and Post-Preliminary processing are not funded here.
 - **Recommended Next Steps:** After KDP2 approval, fund a Phase 5 MAS for Draft FIRM Database and Preliminary Map Production covering Winnebago, Boone, and Ogle Counties. Note this is the highest-dollar active project and involves 3 counties — schedule carefully.
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ISWS 23-09: Alexander and Pulaski County Data Development (Lower Ohio/Cache Watershed)

- **Activity Type:** Data Development (Hydrologic and Hydraulic analysis, 2D and 1D)
 - **Phase:** Data Development through KDP2. Finalizes modeling of Cache River (Pulaski County) and studies initially funded in MAS ISWS18-05 (which was descope to remove regulatory mapping due to ongoing Cache River modeling). Also includes a Flood Risk Review Meeting for Cairo Vicinity Levee gravity drainage study.
 - **Scope:** Completes hydrologic and hydraulic data development for approximately 82 miles of streams in Alexander and Pulaski Counties, using both 1D and 2D modeling. Finalizes Cache River modeling (755 sq mi drainage area) and smaller tributaries (Ketchell, Hodges, Hess Bayou, Cottonwood Slough). Includes community engagement through two Flood Risk Review Meetings (one per county) and coordination with IDNR-OWR for study approval. The Cairo Vicinity levee mapping path (involving Mississippi/Ohio Rivers levee system — currently expired PAL) remains undefined, creating regulatory uncertainty.
 - **Deliverables:** Terrain/topographic data; Hydrology spatial files/reports; Hydraulics simulations/FIS profiles; independent QA/QC reports; Flood Risk Review Meeting documentation; outreach documentation
 - **Period:** 09/01/2023 – 09/30/2026 (terrain data due Sept 2024; all other deliverables Sept 2026)
 - **Budget:** \$395,000 (FEMA only; \$0 partner leverage)
 - **Multi-year:** Yes — this project finishes work originally funded under ISWS18-05 and ongoing Cache River modeling from FY20. Preliminary Map Production, Due Process, and Final FIRM are not funded.
 - **Recommended Next Steps:** Resolve Cairo Vicinity levee mapping path (Mississippi/Ohio River levee system expired PAL) — this is a critical blocker for advancing to regulatory mapping. After KDP2, fund a follow-on MAS for Preliminary Map Production. Consider coordinating with USACE and IDNR-OWR early given the levee complexity in Cairo.
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ISWS 23-10: Embarras River Hydraulics (Lawrence/Crawford/Richland/Jasper/Cumberland/Coles/Douglas/Champaign Counties)

- **Activity Type:** Hydraulic Data Development (mainstem Embarras River only — no floodplain mapping)

- **Phase:** Hydraulics Data Capture only. Prerequisite survey and hydrology were funded under MAS ISWS22-07. This MAS adds the hydraulic analysis layer and does not proceed to floodplain delineation, FIRM Database, or Preliminary production.
 - **Scope:** Completes hydraulic analysis of the Embarras River mainstem from its confluence with the Wabash River upstream to Champaign County — approximately 198 miles (58 miles Zone A approximate, 140 miles Zone AE with floodway). Revises flood zones using updated 1D models. Four non-accredited levee systems exist in the study area with mapping path TBD (requiring future LLPT coordination). The ultimate goal is to replace paper FIRMs in Jasper County with a Digital FIRM.
 - **Deliverables:** Hydraulic simulations; FIS Inserts/Profiles; Floodway Data Tables; spatial files; independent QA/QC; Project Communication Plan; Flood Risk Review Meeting documentation; outreach documentation
 - **Period:** 09/01/2023 – 09/30/2026 (all deliverables due September 2026)
 - **Budget:** \$895,000 (FEMA only; \$0 partner leverage)
 - **Multi-year:** Yes — follows ISWS22-07 (survey and hydrology). Future phases needed: floodplain mapping (Task 16), Draft FIRM Database, Preliminary production, Due Process, Final FIRM. A separate future MAS is also needed for tributary data development before the full Jasper County DFIRM can be produced.
 - **Recommended Next Steps:** Resolve the four TBD levee mapping paths early to avoid delays in future preliminary production. After KDP2, fund floodplain mapping (Task 16) and then a Preliminary production MAS. Also plan a future tributary data development MAS to enable a full Jasper County DFIRM.
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ISWS 23-11: Middle and Lower Wabash Outreach (Bi-State IL/IN Coordination)

- **Activity Type:** Outreach and Coordination (non-technical; no mapping products)
- **Phase:** Post-data development / Pre-regulatory mapping — bridge phase between ISWS18-03 (data development, concluding Aug 2023) and future individual county regulatory mapping projects in Illinois and Indiana
- **Scope:** Provides continued bi-state outreach to stakeholders related to the Middle and Lower Wabash mainstem data development, as individual IL and IN counties begin their own regulatory mapping projects. Key technical deliverable is coordination with Indiana CTP on a master profile and floodway data table for the mainstem Wabash River. Many levee systems on both sides of the river require regional coordination before county-level preliminary maps can be produced.
- **Deliverables:** Project Communication Plan with outreach/coordination documentation; master Wabash River profile and floodway data table (for Indiana CTP); DSC-formatted correspondence and meeting documentation; availability to address technical questions as county mapping projects initiate
- **Period:** 09/01/2023 – 09/30/2026
- **Budget:** \$65,000 (FEMA only; \$0 partner leverage)
- **Multi-year:** Yes — this is a transitional/coordination role between ISWS18-03 and the next generation of county-level Wabash regulatory mapping MAS projects (not yet funded). Eight levee systems documented (most Non-accredited, Natural Valley or TBD mapping path).
- **Recommended Next Steps:** Coordinate with Indiana CTP early to produce the master profile/floodway data. Use this outreach period to scope individual IL county regulatory mapping MAS projects along the Wabash (similar to what was done for Edwards County ISWS 23-06 and Wabash County ISWS 23-07). Engage levee owners (especially Vincennes IN, Brevoort, Mount Carmel) on mapping paths before preliminary production begins.

ISWS 23-12: Henry County Data Development — Green River 2D Study

- **Activity Type:** Data Development (new approximate Zone A, 2D modeling)
- **Phase:** Data Development (~9 miles of Green River between Colona and Geneseo). Extends an upstream Bureau County FY20 MAS 2D model downstream. No floodplain mapping, FIRM Database, or Preliminary production.
- **Scope:** Develops a new 2D hydraulic model (~9 miles) for the Green River between the City of Colona and City of Geneseo in Henry County, extending the upstream Bureau County FY20 2D model to tie into the effective detailed Green River study at Colona (Henry County FIS cross section L). The existing 9-mile Zone A delineation does not reflect observed flooding. Uses 2020 LiDAR (10cm) and Bulletin 17C hydrology (1,063 sq mi drainage area weighted with regression equations).
- **Deliverables:** Terrain data (due March 2025); Hydrology models and spatial files (due June 2026); Hydraulics simulations and spatial files (due August 2026); Event Data Capture/outreach documentation (due September 2026); independent QA/QC reports
- **Period:** 09/01/2023 – 09/30/2026
- **Budget:** \$100,000 (FEMA only; \$0 partner leverage)
- **Multi-year:** Yes — depends on the upstream Bureau County FY20 2D model. Future phases needed: floodplain mapping, FIRM Database, Preliminary production, Due Process, Final Map. Likely to be incorporated into a future Henry County PMR or DFIRM project.
- **Recommended Next Steps:** Coordinate model hand-off timing with the Bureau County FY20 MAS to ensure seamless tie-in. After hydraulic modeling is complete, fund a floodplain mapping (Task 16) effort and then a Henry County PMR/DFIRM MAS to incorporate the new data into regulatory products.

ISWS 23-13: Whiteside County Data Development — Green River Tributaries

- **Activity Type:** Data Development (new approximate Zone A, 2D Rain-on-Mesh)
- **Phase:** Data Development (~10 miles of Main Ditch, Winnebago Ditch, and Winnebago Ditch Tributary A). No floodplain mapping, FIRM Database, or Preliminary production.
- **Scope:** Produces new 2D floodplain studies for three non-model-backed Zone A streams in southeastern Whiteside County (Main Ditch, Winnebago Ditch, Winnebago Ditch Tributary A — all tributaries to the Green River). Uses 2D Rain-on-Mesh (ROM) hydrology/hydraulics methodology across a 58 sq mi domain and 2020 LiDAR (10cm). Explicitly designed to provide improved county edge-matching between Whiteside and Bureau Counties as both move toward future PMR updates.
- **Deliverables:** Terrain data (due March 2025); Hydrology data (2D ROM, due June 2026); Hydraulics simulations/FIS profiles (due August 2026); outreach and Event Data Capture documentation (due September 2026); independent QA/QC reports
- **Period:** 09/01/2023 – 09/30/2026
- **Budget:** \$100,000 (FEMA only; \$0 partner leverage)
- **Multi-year:** Yes — this data feeds a future Whiteside County PMR process (referenced explicitly in the narrative). Bureau County edge-matching dependency. Future phases needed: Floodplain Mapping, FIRM Database, Preliminary production, Due Process, Final Map.

- **Recommended Next Steps:** Coordinate with Bureau County mapping timeline to ensure consistent edge-matching. After hydraulic work is complete, fund a floodplain mapping phase and then a Whiteside County PMR MAS (possibly combined with Bureau County) to incorporate new data into regulatory products.
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ISWS 23-14: Kaskaskia Survey — Bridge and Culvert Data Acquisition

- **Activity Type:** Survey (bridge and culvert data collection)
 - **Phase:** Earliest pipeline stage — pre-engineering survey only. Purely a data acquisition precursor; no H&H modeling, no mapping, no community engagement yet.
 - **Scope:** Collects field survey data (structure dimensions, photos, sketches, as-built data) for bridges and culverts within the Kaskaskia watershed (Upper, Middle, and Lower) in Shelby, Fayette, and Washington Counties. Data is captured and submitted to FEMA's MIP in DSC format to support future H&H data development projects. No communities are formally engaged under this MAS because no regulatory mapping audiences exist yet; communities will be engaged when future H&H development is funded.
 - **Deliverables:** Survey data (photos, sketches, field measurements, as-built data, spatial files) in DSC format — due September 2025; MIP/earned value reporting
 - **Period:** 09/01/2023 – 09/30/2026 (survey deliverable due September 2025 — a full year early)
 - **Budget:** \$200,000 (FEMA only; \$0 partner leverage)
 - **Multi-year:** Yes — this is explicitly an upfront precursor; every downstream step (H&H development, floodplain mapping, FIRM Database, Preliminary, Due Process, Final FIRM) is a future phase requiring separate funding. Zero effective FIRM miles targeted in this MAS.
 - **Recommended Next Steps:** Ensure survey data is submitted to MIP by September 2025 as scheduled. Use ISWS 23-02 Kaskaskia strategic planning outputs (watershed prioritization) to determine which specific Kaskaskia sub-reaches should be targeted for the first H&H data development MAS. Fund a Kaskaskia H&H Data Development MAS in FFY2024 or FFY2025 to begin using this survey data.
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Cross-Cutting Observations

1. All Production MAS Projects Stop at Preliminary — Due Process Phase Is a Funding Gap

Five regulatory mapping/production projects (ISWS 23-03 through 23-07) all explicitly exclude Due Process and Final Map Production tasks. This means that after completing Open Houses and CCO meetings, these projects will stall without a follow-on funded MAS to complete the regulatory cycle (appeal period resolution, LFD issuance, final FIRM printing). **All five need follow-on FFY2024 or FFY2025 MAS funding for Due Process + Final Map Production.** McDonough County (23-05, already at QR2) is likely the nearest-term candidate.

2. Edwards County and Wabash County Must Be Managed in Tandem

ISWS 23-06 and ISWS 23-07 share a common risk: Bonpas Creek forms the Edwards/Wabash county boundary, and any substantial comment on that reach will affect both projects' budgets and schedules. These two projects should be funded, coordinated, and advanced through Due Process simultaneously.

3. The Kishwaukee Watershed Is the Single Largest Investment — and Still Far from Final FIRM

At \$1.697M, ISWS 23-08 (Phase 4 Data Development) represents 35% of the entire FFY2023 portfolio budget. The Kishwaukee multi-phase effort spans DeKalb, McHenry, Winnebago, Boone, and Ogle Counties. Phase 4 only reaches KDP2 (data development). Phase 5 (Preliminary production) and eventually Phase 6 (Due Process/Final FIRM) still need to be scoped and funded. The 3-county Phase 4 footprint is large — schedule Phase 5 carefully.

4. The Embarras River and Kaskaskia Are Long-Runway Pipeline Projects

ISWS 23-10 (Embarras River Hydraulics, \$895K) is the hydraulic analysis step only; it follows survey/hydrology work from ISWS 22-07 and still requires floodplain mapping, Preliminary, and Final FIRM steps — plus tributary data development — before a Jasper County DFIRM can be produced. ISWS 23-14 (Kaskaskia Survey, \$200K) is even earlier in the pipeline. Both are 4–6 year minimum pipelines from current state to final effective FIRM.

5. The Wabash River Corridor Appears Repeatedly

Wabash-related projects appear across four distinct MAS entries: ISWS 23-07 (Wabash County PMR), ISWS 23-09 (Alexander/Pulaski — Lower Ohio/Cache, Wabash confluence area), ISWS 23-10 (Embarras River confluence with Wabash), and ISWS 23-11 (Middle/Lower Wabash bi-state outreach). The Mt. Carmel levee (Wabash County) and Cairo levee system (Alexander County) both have unresolved accreditation/mapping paths that represent scheduling risk for advancing these projects.

6. Green River / Bureau County Dependency Appears Twice

Both ISWS 23-12 (Henry County) and ISWS 23-13 (Whiteside County) reference the Bureau County FY20 upstream model as a dependency. These three projects collectively form a connected Green River data pipeline. Consider sequencing or bundling a future PMR/DFIRM MAS that covers all three counties' new data in a single production phase.

7. Kaskaskia Watershed Has Two Simultaneous Touchpoints

ISWS 23-02 (COMS) focuses on Kaskaskia strategic planning and community prioritization for future mapping, while ISWS 23-14 launches the survey data pipeline for Kaskaskia H&H development. Coordinating the prioritization output from 23-02 with the target stream selection for 23-14 and future H&H MAS is essential to ensure survey data collected matches where modeling effort will ultimately go.

8. Budget Concentration Risk

Three projects (ISWS 23-08 Kishwaukee: \$1.697M; ISWS 23-10 Embarras: \$895K; ISWS 23-09 Alexander/Pulaski: \$395K) account for approximately 61% of total FFY2023 project funding. These are all data development projects that have not yet produced a regulatory product. Performance risk management is important for these three.

9. All Projects Have Zero Partner Leverage

Every project in the FFY2023 portfolio lists \$0 partner contribution/leverage. This is consistent with ISWS's CTP model but worth noting — there may be opportunities in future cycles to document in-kind contributions from IDNR-OWR or county partners.

10. FFY2023 Portfolio: Summary Budget Table

MAS No.	Project	Type	Budget
ISWS 23-01	Program Management	PM	\$95,000
ISWS 23-02	Community Outreach & Mitigation (Kaskaskia)	COMS	\$200,000
ISWS 23-02A	COMS Amendment (language only)	Amendment	\$0 additional
ISWS 23-03	Peoria County DFIRM	Regulatory Mapping	\$260,100
ISWS 23-04	McHenry/Nippersink PMR	Regulatory Mapping	\$252,100
ISWS 23-05	McDonough County DFIRM	Regulatory Mapping	\$206,360
ISWS 23-06	Edwards County DFIRM	Regulatory Mapping	\$205,500
ISWS 23-07	Wabash County PMR	Regulatory Mapping	\$237,540
ISWS 23-08	Kishwaukee Phase 4 DD	Data Development	\$1,697,002
ISWS 23-09	Alexander/Pulaski DD	Data Development	\$395,000
ISWS 23-10	Embarras River Hydraulics	Data Development	\$895,000
ISWS 23-11	Mid/Lower Wabash Outreach	Outreach/Coordination	\$65,000
ISWS 23-12	Henry County DD (Green River)	Data Development	\$100,000
ISWS 23-13	Whiteside County DD	Data Development	\$100,000
ISWS 23-14	Kaskaskia Survey	Survey	\$200,000
TOTAL			\$4,908,602